

Self Driving Car

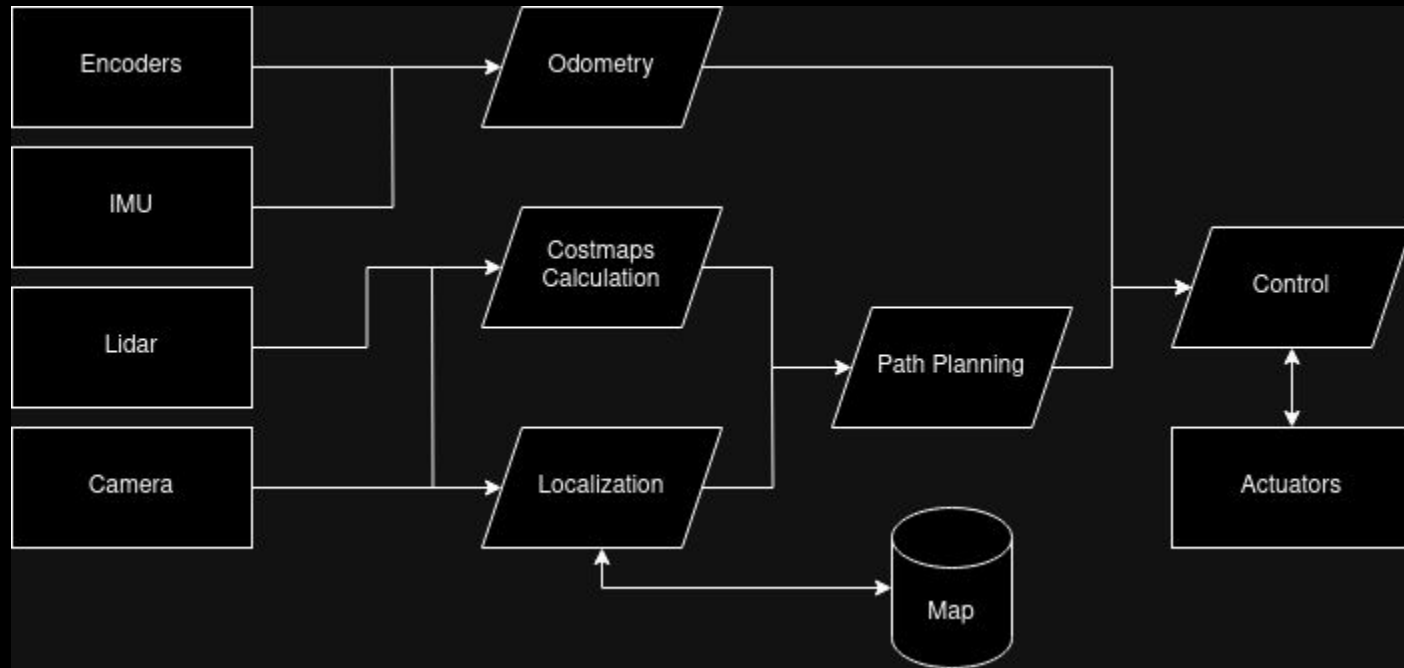
A Project Progress Presentation

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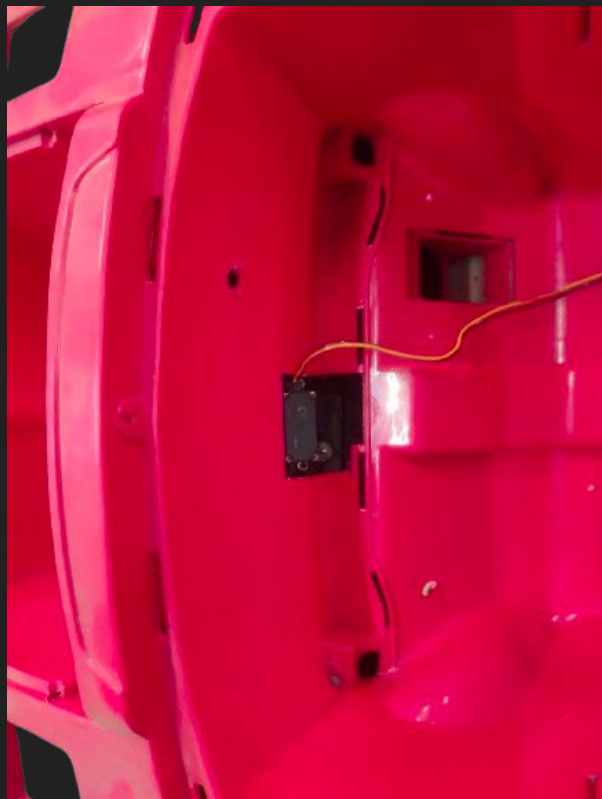
Project Overview

- Why
 - Cars need to be safer
 - Home-grown technology is crucial
- Goal - Self driving car using commodity hardware
- Scope
 - Autonomy within the western campus compound (ICTC to Arch. dept)
 - Use ROS2 for middleware
 - Reuse components from Nav2
 - Target slow speed regimes in good lighting conditions

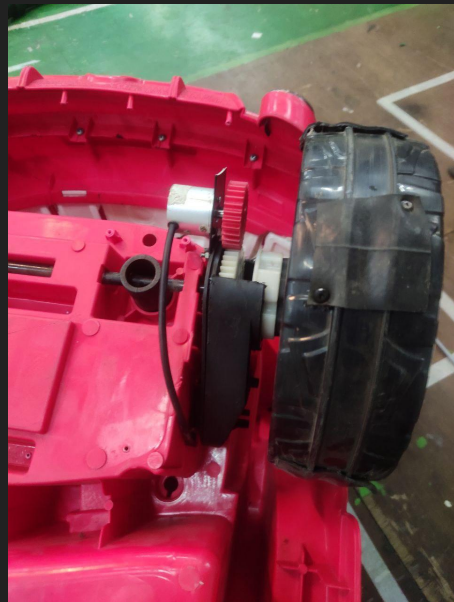
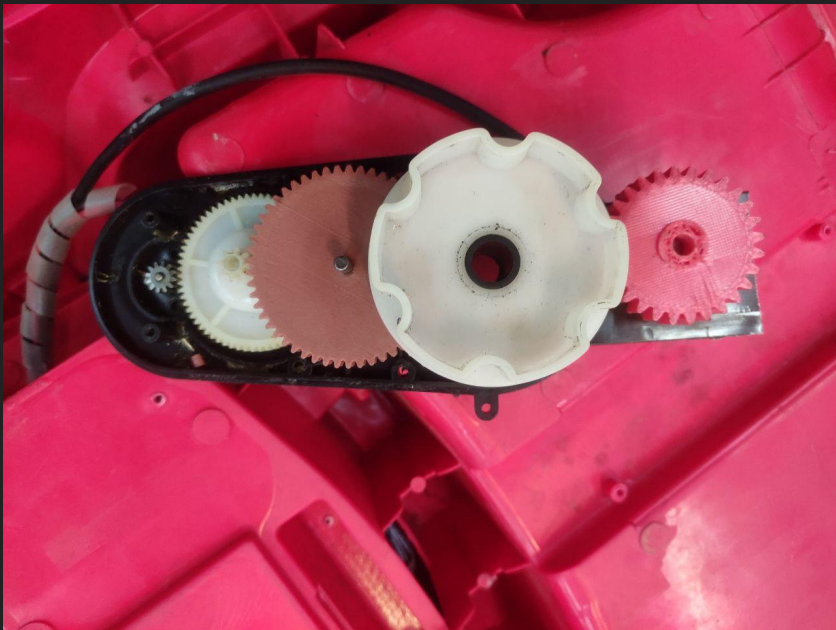
System Design



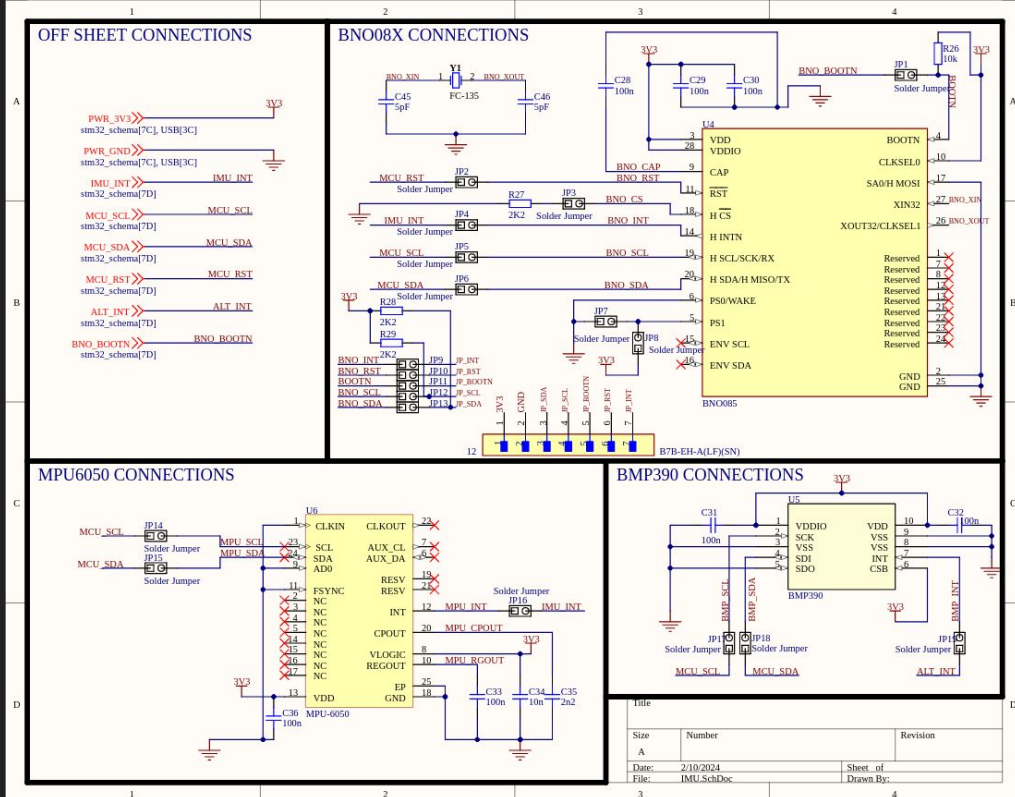
Task Completed - Hardware



Task Completed - Hardware



Task Completed - Hardware



Task Completed - Hardware



Task Completed - Software

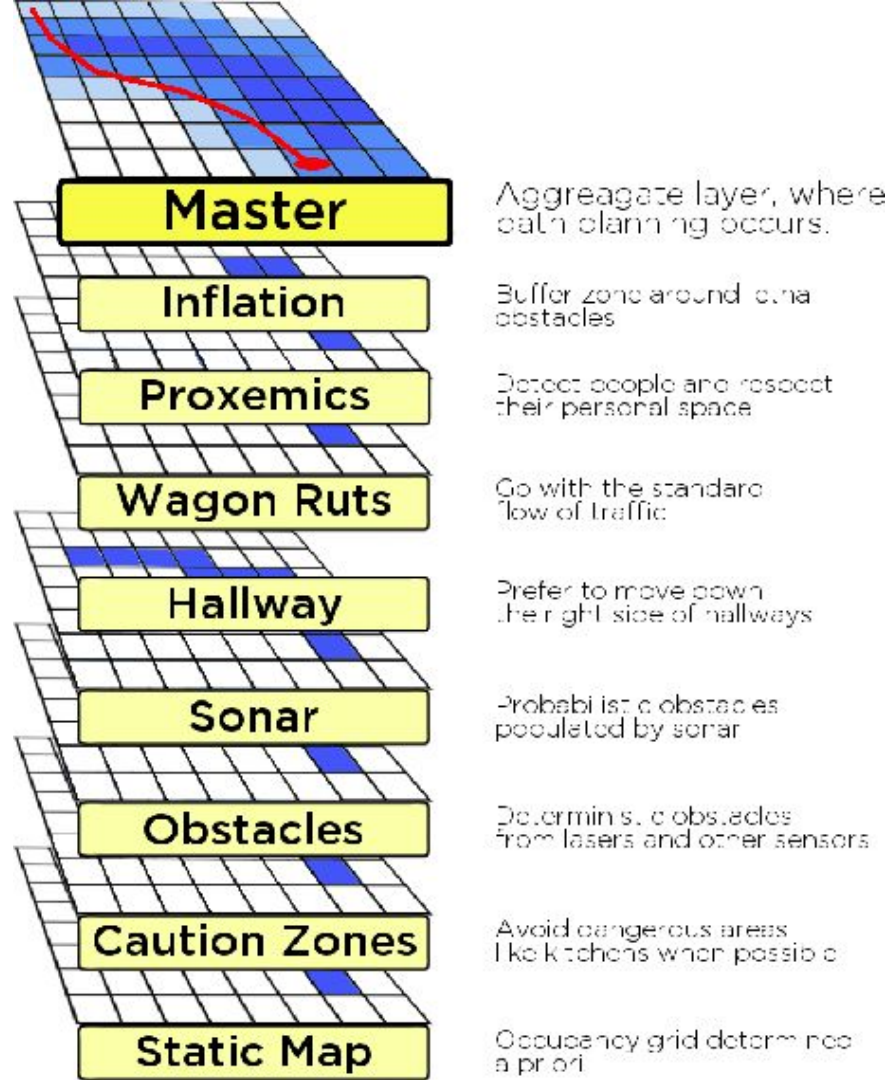
- Experiments with car geometry - Ackermann and Bicycle models
- ROS2 Control - Tests in simulation, Hardware Interface setup
- SLAM experiments
 - Lidar slam with SLAM Toolbox
 - Monocular SLAM with Stella VSLAM
- Motor speed control with PID controller
- Forward kinematics on the microcontroller

Task Completed - Software



Task Remaining

- Mapping
 - Visual+2D Lidar metric map
- Odometry
 - Fusion of odometry data from 4 sources
- Costmap factors and fusion
 - Obstacle avoidance + road keeping + lane preferences
- Perception Models
 - Tune obstacle detection and drivable region segmentation
- Navigation testing and tuning
 - End to end tests of planners, controllers, behaviour trees



THANK YOU